

Advancing Financial Engineering with Foundation Models: Progress, Applications, and Challenges

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ABSTRACT

The advent of foundation models (FMs)—large-scale pre-trained models with strong generalization capabilities—has opened new frontiers for financial engineering. While general-purpose FMs such as GPT-4 and Gemini have demonstrated promising performance in tasks ranging from financial report summarization to sentiment-aware forecasting, many financial applications remain constrained by unique domain requirements such as multimodal reasoning, regulatory compliance, and data privacy. These challenges have spurred the emergence of **Financial Foundation Models (FFMs)**—a new class of models explicitly designed for finance. This survey presents a comprehensive overview of FFMs, with a taxonomy spanning three key modalities: Financial Language Foundation Models (FinLFMs), Financial Time-Series Foundation Models (FinTSFMs), and Financial Visual-Language Foundation Models (FinVLFMs). We review their architectures, training methodologies, datasets, and real-world applications. Furthermore, we identify critical challenges in data availability, algorithmic scalability, and infrastructure constraints, and offer insights into future research opportunities. We hope this survey serves as both a comprehensive reference for understanding FFMs and a practical roadmap for future innovation. A updated collection of FFM-related publications and resources will be maintained on our website <https://github.com/FinFM/Awesome-FinFMs>.

1. Introduction

Financial engineering, an interdisciplinary domain that combines finance, mathematics, and computer science, has become the cornerstone of modern financial systems. It plays a critical role in designing complex financial products, managing risk, and supporting decision-making through quantitative models and data-driven algorithms [1]. As financial markets grow in complexity and data heterogeneity, there is an increasing need for intelligent systems [2, 3, 4, 5] that can generalize across a wide range of tasks, adapt to changing environments, and support scalable, automated financial workflows.

Recent advancements in **Foundation Models (FMs)**, i.e., large-scale pre-trained models such as GPT-4 [6], Gemini [7], Qwen [8] and DeepSeek [9], have demonstrated impressive generalization capabilities across modalities and domains. In financial contexts, these general-purpose FMs have already begun to reshape traditional pipelines, offering strong performance on tasks such as financial report summarization, portfolio commentary generation, sentiment-aware forecasting, and risk disclosure analysis [10, 11, 12]. This emerging paradigm shift—from narrowly specialized AI systems to general-purpose foundation models—promises to transform the landscape of financial engineering, as illustrated in Figure 1.

Despite the strong potential of general-purpose FMs in finance, many financial tasks possess unique characteristics—such as legal compliance, multimodal document processing, long-horizon time-series analysis, and strict privacy requirements—that are not fully addressed by off-the-shelf models. These challenges have motivated the research of **Financial Foundation Models (FFMs)**—a class of FMs designed specifically for financial scenarios through domain-aware pretraining, task-specific fine-tuning, and alignment with financial reasoning and regulatory goals.

In this survey, our aim is to systematize the development of FFMs. We categorize FFMs into three major types based on input modality and application scope:

- **Financial Language Foundation Models (FinLFMs):** Also referred to as Financial Large Language Models (FinLLMs) [13, 14], FinLFMs are models pre-trained on financial texts, including reports, news, and contracts, optimized for tasks like question answering, summarization, and compliance checking [15, 16].
- **Financial Time-Series Foundation Models (FinTSFMs):** FinTSFMs extend FFMs beyond language by processing sequential financial data (e.g., price histories, economic indicators) for financial time-series data analysis tasks such as stock price forecasting, volatility modeling, and risk recognition [17].
- **Financial Visual-Language Foundation Models (FinVLFMs):** FinVLFMs are designed to process both textual and visual information, such as financial charts,

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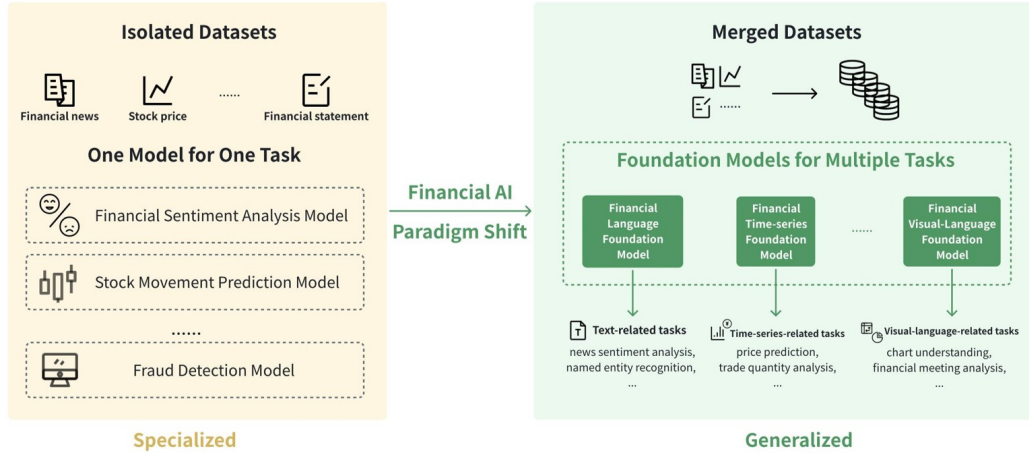


Figure 1: Financial AI paradigm shifts from traditional models for specific tasks to foundation models for multiple tasks.

tables, and figures [18, 19, 20], enabling complex tasks that require multimodal understanding.

These three categories collectively represent the ongoing shift from narrow task-specific modeling to broadly adaptable and general-purpose systems, enabling a new era of scalable, multimodal, and intelligent financial engineering [21, 16, 10]. In this survey, we aim to explore the current state of FFMs comprehensively. The following questions guide our study: 1) Despite the remarkable achievements of foundation models in other fields, what are the current progresses of FFMs in the financial industry? 2) With the development of these models, what challenges do they encounter in terms of data, algorithms, and infrastructure? By answering these questions, we hope to provide a comprehensive overview of the current situation of FFMs and offer a clear vision for their future development. In this article, we focus on the current progress of FinLFMs, FinTSFMs and FinVLFMs from 2018 (the start of the foundation model era in a broader sense [13]) to 2025 June, along with their challenges and opportunities. We believe this survey will assist financial researchers and practitioners in quickly grasping the development of FFMs and inspire new ideas for further innovation.

1.1. Brief History of Foundation Models in Finance

The evolution of Financial Foundation Models (FFMs) mirrors key milestones in the broader development of AI. The launch of BERT [22] in 2018 marked the start of the pretraining era, leading to early finance-specific models such as FinBERT [23, 24, 25] tailored for financial text understanding. This initiated the first wave of FinLFMs.

The emergence of GPT-3 and ChatGPT, built on the now-famous scaling law [26], demonstrated the power of massive autoregressive models and instruction tuning. These advances inspired a new generation of FinLFMs that leveraged generative capabilities, human feedback alignment, and multi-task transfer [13, 14]. However, such models struggled

with non-textual inputs, particularly the sequential structure of financial markets.

To address this limitation, FinTSFMs [27] emerged, either by directly pretraining on historical price data or by adapting LLMs through hybrid designs. Some even blended language and numerical inputs, reflecting a shift toward modality fusion.

More recently, models like GPT-4o [28] have expanded foundation models into the multimodal domain. This inspired the development of FinVLFMs [18, 19, 20], capable of jointly processing financial charts, tables, and text. These models aim to provide a more integrated and human-like understanding of financial scenarios.

Together, the progression from FinLFMs to FinTSFMs and FinVLFMs marks the trajectory of FFMs toward broader generality, stronger reasoning, and richer modality integration in financial AI.

1.2. Comparison of Related Surveys and Our Contributions

In our extensive literature search, we identified nine representative surveys on FFMs. These existing works have provided valuable insights from various perspectives. However, compared to prior studies, our survey offers a more comprehensive and in-depth examination of FFMs—spanning methodologies, data sources, applications, as well as detailed discussions on current challenges and future research directions. Table 1 summarizes the key distinctions between our survey and existing literature. Specifically, our survey presents the following unique contributions:

1. **A systematic taxonomy and thorough coverage of FFM sub-fields.** Our survey categorizes FFMs into three major sub-fields—language, time-series, and visual-language models—in the context of financial tasks. This comprehensive taxonomy provides a unified view of the entire FFM landscape, surpassing the scope of existing surveys that tend to focus on specific modalities.

Survey	FinLFMs	FinTSFMs	FinVLFMs	Datasets	Applications	Challenges
[21]	◦	×	×	×	◦	◦
[29]	×	×	×	×	✓	◦
[16]	×	×	×	✓	✓	◦
[10]	✓	×	×	✓	✓	✓
[11]	◦	×	×	◦	◦	◦
[12]	×	×	×	✓	◦	◦
[30]	×	×	×	×	◦	◦
[31]	×	×	×	×	✓	✓
[32]	◦	×	×	◦	◦	◦
Our Survey	✓	✓	✓	✓	✓	✓

Table 1

Comparison between our survey and existing surveys on Financial Foundation Models. FinLFMs, FinTSFMs, and FinVLFMs stand for financial Language/Time-Series/Visual-Language foundation models, respectively. Note: × indicates no content discussion, ◦ indicates brief and non-comprehensive discussion, and ✓ indicates comprehensive discussion.

2. **A holistic and deeper exploration of multiple dimensions of FFMs.** We investigate FFMs from a range of perspectives, including modeling techniques, data characteristics, practical applications, technical challenges, and potential future directions. This integrated approach allows readers to gain a more complete and nuanced understanding of FFMs compared to previous reviews.
3. **An up-to-date and practitioner-oriented synthesis of methods and resources.** Beyond reviewing published work, we provide a curated and continuously updated repository of FFM-related papers, tools, and datasets on our companion website <https://github.com/FinFM/Awesome-FinFFMs>. This living resource serves as a practical reference hub for researchers and practitioners, which is rarely addressed in other surveys.

Our contributions in this survey are as follows:

1. **Systematic Review of Methods (Sections 2–4):** We propose a novel taxonomy of Financial Foundation Models and conduct a comprehensive review of existing methods across the three major sub-fields: language (Section 2), time-series (Section 3), and visual-language (Section 4). We also investigate different training strategies, offering insights into potential technical innovations for FFMs.
2. **Comprehensive Survey of Datasets (Sections 2–4):** Alongside the model review, we survey a wide range of large-scale datasets and databases that are applicable to training FFMs in different modalities. This survey helps identify key limitations of current financial data resources and provides practical guidance for dataset selection in future research.
3. **Thorough Overview of Applications (Section 5):** We present an overview of diverse financial applications enabled by FFMs, illustrating how these models are currently being deployed in real-world financial engineering tasks. This serves as a reference point for extending FFMs to new application scenarios.

4. **In-depth Discussion of Key Challenges and Future Research (Section 6):** We analyze critical challenges related to data, algorithm, and computing infrastructure in the development and deployment of FFMs. These discussions identify major bottlenecks and open new directions for future research.

2. Financial Language Foundation Models

To begin with, we first introduce Financial Language Foundation Models (FinLFMs) in this section. FinLFMs represent a class of domain-specific LFMs that are specifically trained on financial corpora to acquire text comprehension, reasoning, and generation capabilities tailored for financial scenarios. We investigate FinLFMs through three analytical lenses: (1) Collection and Category: This part describes how the reviewed FinLFMs are collected and classified into Bert-style FinLFMs, GPT-style FinLFMs, and reasoning-enhanced FinLFMs. (2) Training Method Analysis: In this part, we delve into the innovations across pre-training strategies, supervised fine-tuning techniques, and alignment mechanisms for training FinLFMs. (3) Language-related Datasets: This part introduces the commonly used datasets in the training and evaluation of FinLFMs.

2.1. Model Classification

To comprehensively review FinLFMs, we conducted an extensive search across multiple channels (e.g., Google Scholar, Web of Science, ArXiv) and scoured GitHub repositories and official announcements from major financial institutions and tech companies for open-source and proprietary FinLFM projects. In total, our survey covers 21 representative FinLFM models.

The evolution of FinLFMs closely mirrors that of general LFMs. Fig 2 shows the development trajectory of reviewed FinLFMs based on their architectural backbones with timeline. Guided by this timeline, we categorize FinLFMs into three groups: Bert-style, GPT-style, and reasoning-enhanced FinLFMs.

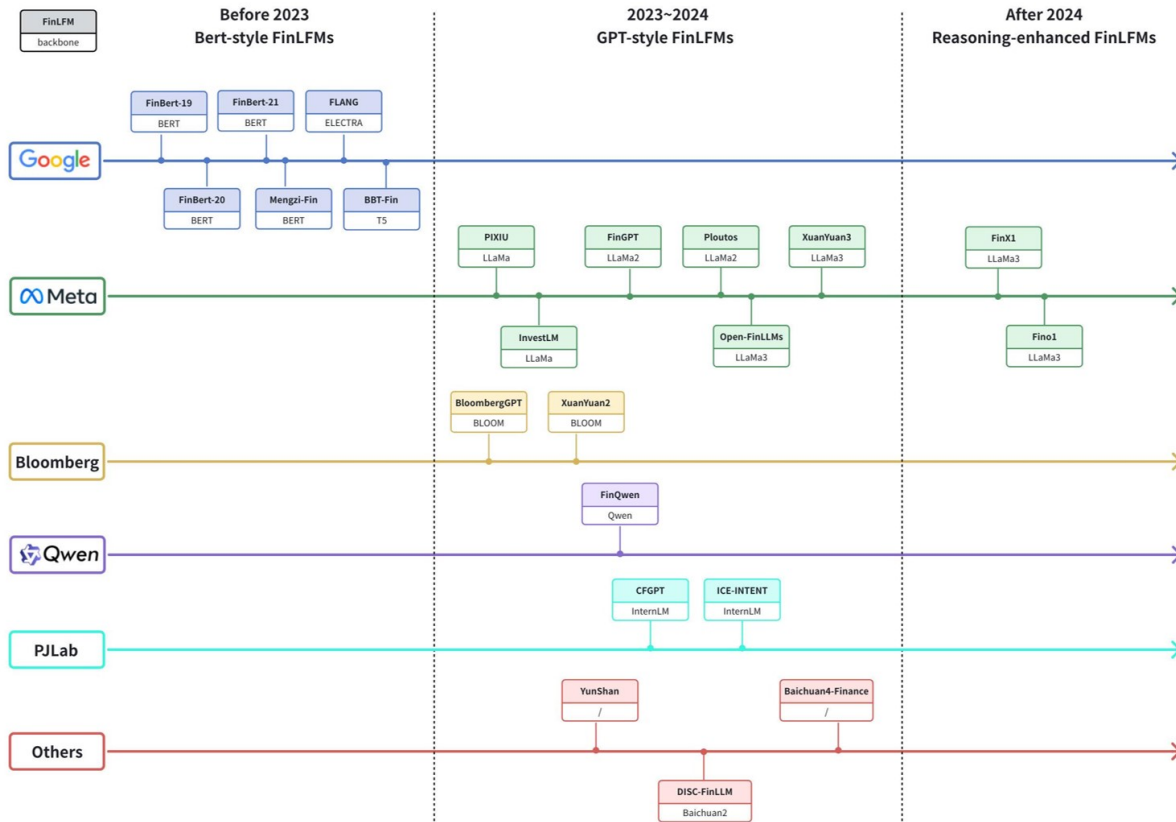


Figure 2: The development trajectory of Financial Language Foundation Models (FinLFMs) based on their architectural backbones with timeline.

2.1.1. Bert-style FinLFMs

The first generation of finance-specific FinLFMs adapted Bert-style LMs to financial text. For instance, FinBERT-19 [23] continued pre-training BERT on financial news. At the same time, there are also FinBERT-20 [24] and FinBERT-21 [25], which were further pre-trained on different and mixed general and financial corpora. Additionally, [33] explore to obtain a Chinese financial BERT-style model named as Mengzi-fin. Another example is FLANG [34], a finance-specialized variant of ELECTRA [35] that uses selective masking tailored to financial keywords and phrases for better performance. More recently, BBT-Fin [36] introduced a T5-style[37] financial PLM with knowledge-enhanced pretraining. Together, these Bert-style FinLFMs learned domain-specific vocabularies and patterns, forming the backbone for many later systems.

2.1.2. GPT-style FinLFMs

The next wave of FinLFMs was catalyzed by the success of ChatGPT [38], with BloombergGPT [15] marking a pivotal start. Built on the BLOOM [39] architecture and trained on Bloomberg’s proprietary data, BloombergGPT demonstrated the feasibility of domain-specific LLMs at scale. Since then, numerous finance-oriented variants have emerged, including XuanYuan2 [40] for Chinese markets. Crucially, the rise of GPT-style FinLFMs has paralleled the rapid evolution of their open-source foundations. As

general-purpose LLMs like LLaMA [41, 42] Qwen [8], and InternLM [43] continue to scale, FinLFMs have inherited architectural advancements—e.g., deeper attention, longer context, and stronger instruction following—while layering in financial expertise via supervised finetuning or continual pretraining. This symbiotic relationship enables fast capability upgrades without training from scratch. For instance, Meta’s LLaMA series has underpinned models like PIXIU [44], InvestLM [45], FinGPT [46, 47], Open-FinLLMs [19], Ploutos [48], and XuanYuan3 [49], while Qwen inspired Qwen-Fin [50]. InternLM, another open-source backbone proposed by PjLab, powers large-scale models like ICE-INTENT [51] and CFGPT [52], which combine extensive instruction datasets with financial domain knowledge. Collectively, these developments highlight how FinLFMs increasingly benefit from and contribute to the open-source LLM ecosystem. There also exist some commercial models, such as YunShan [53] and Baichuan4-Finance [54], follow similar patterns without disclosing full architectural details.

2.1.3. Reasoning-enhanced FinLFMs

More recently, a new class of FinLFMs has emerged with advanced reasoning and decision-making capabilities following the success of OpenAI-o1 and Deepseek-R1 [55], which enhance general LLMs with chain-of-thought (CoT) prompting. For instance, Fin-X1 [49] and Fin-o1[56] use

Chain-of-Thoughts to guide the models to generate intermediate reasoning steps before arriving at the final answer. These models represent a transition toward autonomous financial agents, built not only to respond but to reason, plan, and act across complex tasks with step-by-step CoTs.

2.2. Training Method Analysis

Consequently, this section focuses on GPT-style FinLFMs, which now dominate the field in both research and deployment. Their training strategies are summarized in Table 2, following the standard three-stage framework—pre-training, supervised finetuning, and alignment—proposed in the State of GPT report by Karpathy [57].

2.2.1. Pre-training (PT)

Pre-training establishes the core language understanding and reasoning capabilities of FinLFMs. While early models like BloombergGPT [15] were trained from scratch on massive financial corpora, recent trends favor continuous pre-training (CPT) on top of open-source general-purpose LLMs (e.g., LLaMA, Qwen), combining general and financial domains. This strategy, employed by models such as FinQwen [50] and Open-FinLLMs [19], enhances domain specialization without sacrificing generalization. Pre-training typically involves more than 100 billion tokens and requires large-scale distributed infrastructure, thus remaining mostly within the scope of industry labs. For instance, Baichuan4-Finance [54] employs 500B tokens across general and financial domains. In contrast, academic models often skip this stage due to resource limitations, relying instead on downstream fine-tuning, which is introduced next.

2.2.2. Supervised Finetuning (SFT)

Supervised finetuning adapts the pre-trained model to financial downstream tasks using high-quality instruction datasets. A distinguishing feature of FinLFMs is the emphasis on instruction-following formats tailored to financial contexts—e.g., forecasting, question answering, risk assessment, and summarization. Recent models such as FinGPT [46, 47], PIXIU [44], and Ploutos [48] leverage instruction datasets ranging from 30K to 150K examples. High-quality curation is critical: InvestLM [45], for example, demonstrates satisfied performance with only 1335 expert-crafted instructions. CFGPT [52] represents one of the most expansive efforts, using over 1.5 million financial instructions in addition to 141 billion tokens during pretraining. ICE-INTENT [51] also demonstrates the potential of multilingual and multi-dataset fusion by integrating over 600K bilingual financial instructions.

2.2.3. Alignment

Alignment constitutes the final stage, where FinLFMs are optimized to meet domain-specific behavioral objectives—factual accuracy, regulatory consistency, and reasoning transparency. Two major trends have emerged in alignment for financial models: (1) Compliance-focused alignment, as seen in XuanYuan3 [49] and Baichuan4-Finance [54], employs reward models and reinforcement

learning (e.g., Proximal Policy Optimization, PPO) to penalize hallucinations and ensure output consistency with legal and financial norms. (2) Reasoning-centric alignment, exemplified by FinX1 and Fino1 [56], uses techniques such as Generalized Rejection Policy Optimization (GRPO) to encourage detailed, interpretable chain-of-thought reasoning for high-stakes tasks. Despite its relatively nascent development, alignment is poised to become a cornerstone of future FinLFM research. As financial applications demand increasing levels of trustworthiness, interpretability, and regulatory compliance, alignment techniques offer a promising pathway for bridging model capabilities with domain-specific behavioral expectations. Future efforts will likely focus on developing standardized evaluation benchmarks, more robust reward modeling frameworks, and scalable alignment pipelines tailored to the financial domain.

2.3. Financial text-based Datasets

In this part, we summarize representative financial text-based datasets in Table 3. These datasets that have been developed to support the training and evaluation of FinLFMs. These datasets span a variety of tasks, languages, and modalities, reflecting the growing maturity and diversity of financial NLP resources. We categorize the evolution of financial-text datasets into three stages based on their scale, scope, and language coverage.

2.3.1. Early Stage: Task-specific and English-centric

Initial financial NLP datasets were typically small-scale, focused on a single task, and exclusively in English. For example, the Financial PhraseBank (FPB) [59] provided sentence-level annotations for sentiment analysis; FiQA [60] introduced fine-grained annotations across microblogs and headlines for sentiment and Question-Answering (QA) tasks; and ACL18 [61], along with CIKM18 [62], collected tweets for stock movement prediction. These datasets enabled early FinLFMs such as FinBERT-19 [23], FLANG [34], and BloombergGPT [15] to demonstrate domain adaptation and transfer learning capabilities.

2.3.2. Mid Stage: Multi-task Integration and Language Expansion

The next wave of dataset development involved integrating multiple financial NLP tasks and expanding linguistic diversity. FLUE [34] and FLARE [44] consolidated earlier benchmarks into unified evaluation suites, covering sentiment analysis, QA, Named Entity Recognition, and forecasting. Parallel efforts in the Chinese financial domain resulted in datasets such as FinEval [63], CFBenchmark [64], FinanceIQ [65], CFinBench [66], and FLAME [67], which simulate certification exams and textbook-style reasoning tasks. These multilingual, multi-task resources have significantly broadened the benchmarking landscape for non-English FinLFMs.

Table 2
Training Details of Representative FinLFMs

Model	Backbone	Method	Parameters	Additional Training Data	Reference	Open Source
BloombergGPT	BLOOM	PT, SFT	50B	[G]345B+[F]363B tokens	[15]	No
PIXIU	LLaMA	SFT	7B, 30B	[F]136K instructions	[44]	Yes ¹
InvestLM	LLaMA	SFT	65B	[F]1335 instructions	[45]	Yes ²
XuanYuan2	BLOOM	PT, SFT	176B	/	[40]	Yes ³
YunShan	/	PT, SFT	7B	[F]36.5B tokens	[53]	No
FinQwen	Qwen	PT, SFT	14B	[F]200B tokens	[50]	Yes ⁴
FinGPT	LLaMA2	SFT	7B	[F]151K instructions	[47]	Yes ⁵
Ploutos	LLaMA2	SFT	7B	[F]~30K instructions	[48]	No
DISC-FinLLM	Baichuan2	SFT	/	[F]250K instructions	[58]	Yes ⁶
CFGPT	InternLM	PT, SFT	7B	[F]141B tokens + 1.5M instructions	[52]	Yes ⁷
ICE-INTENT	InternLM	SFT	7B	[F]604K bilingual samples from 36 datasets	[51]	Yes ⁸
Open-FinLLMs	LLaMA3	PT, SFT	8B	[G]18B + [F]52B tokens	[19]	Yes ⁹
XuanYuan3	LLaMA3	PT, SFT, Alignment	70B	([G]+[F])275B tokens	[49]	Yes ³
Baichuan4-Finance	/	PT, SFT, Alignment	/	[G]400B+[F]100B tokens	[54]	No
FinX1	LLaMA3	SFT, Alignment	70B	/	[49]	Yes ³
Fino1	LLaMA3	SFT, Alignment	8B	[F]5000 instructions	[56]	Yes ¹⁰

[PT]=Pre-training, [SFT]=Supervised Finetuning, [G]=General Corpus, [F]=Financial Corpus

¹ <https://github.com/The-FinAI/PIXIU>

² <https://github.com/AbaciNLP/InvestLM>

³ <https://github.com/Duxiaoman-DI/XuanYuan>

⁴ <https://github.com/Tongyi-EconML/FinQwen>

⁵ <https://github.com/AI4Finance-Foundation/FinGPT>

⁶ <https://github.com/FudanDISC/DISC-FinLLM>

⁷ <https://github.com/TongjiFinLab/CFGPT>

⁸ <https://github.com/YY0649/ICE-PIXIU>

⁹ <https://anonymous.4open.science/r/PIXIU2-0D70/B1D7/LICENSE>

¹⁰ <https://github.com/The-FinAI/Fino1>

2.3.3. Recent Stage: Cross-lingual and Real-world Benchmarks

More recent datasets push toward cross-lingual, instruction-tuned, and real-world financial understanding. Datasets such as ICE-FLARE [51], CFLUE [68], and FinBen [69] offer multi-task evaluation across English and Chinese in retrieval, QA, and classification settings. AlphaFin [70] integrates large-scale CoT-enhanced QA samples with document retrieval, targeting complex financial reasoning. Meanwhile, M³FinMeeting [71] and FinGEITje [72] introduce multilingual financial dialogues, meeting summaries, and Dutch financial QA tasks—emphasizing realistic, interactive applications. These resources mark a shift toward comprehensive, practical benchmarks that reflect the practical requirements of advanced FinLFMs.

2.4. Summary

FinLFMs have evolved rapidly from BERT-style encoders to GPT-style chatbots and reasoning-enhanced agents. This evolution has been driven by advances in updated and open-source base LFMs, scalable training pipelines, and increasingly diverse financial datasets. While FinLFMs now

serve as the backbone for a wide range of financial NLP tasks, their limitations in handling temporal dynamics and multimodal content underscore the need for foundation models that go beyond language. In the following sections, we turn our attention to FinTSFMs and FinVLFMs, which aim to complement FinLFMs and build more holistic financial AI systems.

3. Financial Time-Series Foundation Models

This section introduces Financial Time-Series Foundation Models (FinTSFMs), which are developed to analyze financial time-series data, including stock price, order book flow, etc. Similar to FinLFMs, we categorize FinTSFMs from three analytical perspectives: (1) collection and category, (2) training strategies, and (3) financial time-series datasets.

3.1. Model Classification

While FinLFMs have rapidly matured over recent years, the exploration of FinTSFMs remains at an early stage. In this section, we aim to provide a representative overview of existing FinTSFM efforts. Through an extensive literature

Table 3
Common Financial-text Datasets

Dataset	Task(s)	Language	Scale	Open Source
FPB [59]	Sentiment Analysis	English	4,840 financial news sentences	Yes ¹
NER [73]	Named Entity Recognition	English	1,473 annotated sentences from 8 financial agreements	Yes ²
FiQA [60]	Sentiment Analysis Question Answering	English	675 microblogs + 498 headlines	Yes ³
ACL18 [61]	Stock Prediction	English	Tweets and daily prices for 88 stocks (2014–2016)	Yes ⁴
CIKM18 [62]	Stock Prediction	English	746,287 tweets for 47 stocks over 231 days (2017)	Yes ⁵
Headline [74]	News Classification	English	11,412 financial news headlines	Yes ⁶
ConvFinQA [75]	Question Answering	English	3,892 dialogues with 14,115 Question Answering pairs	Yes ⁷
FLUE [34]	Multi-task Evaluation	English	Aggregates FPB, FiQA, NER, Headline, ConvFinQA	Yes ⁸
FLARE [44]	Multi-task Evaluation	English	Unified benchmark combining 9+ FinNLP datasets	Yes ⁹
FinEval [63]	Multi-task Evaluation	Chinese	8,351 Question Answering samples from exams, textbooks, and websites	Yes ¹⁰
CFBenchmark [64]	Multi-task Evaluation	Chinese	3,917 financial texts across 8 tasks for recognition, classification, generation	Yes ¹¹
FinancelQ [65]	Multi-task Evaluation	Chinese	7,173 multiple-choice questions across 10 topics	Yes ¹²
FinBen [69]	Multi-task Evaluation	English	36 datasets covering 24 financial NLP tasks	Yes ⁹
AlphaFin [70]	Multi-task Evaluation	English	~220K Question Answering samples (stock/news/report) with retrieval + chain-of-thought	Yes ¹³
ICE-FLARE [51]	Cross-lingual Multi-task Evaluation	English Chinese	21 tasks covering Question Answering and classification	Yes ¹⁴
CFinBench [66]	Multi-task Evaluation	Chinese	99,100 mock exam questions across law, finance, practice	Yes ¹⁵
FinGEITje [72]	Cross-lingual Multi-task Evaluation	Dutch English	140K instruction-tuning samples + Dutch Question Answering benchmark	Yes ¹⁶
CFLUE [68]	Multi-task Evaluation	Chinese	>38K multiple-choice questions + 16K application tests	Yes ¹⁷
M ³ FinMeeting [71]	Cross-lingual Multi-task Evaluation	English Chinese Japanese	Meeting dialogues in 11 sectors with summarization and Question Answering pairs	Yes ¹⁸
FLAME [67]	Multi-task Evaluation	Chinese	16K exam questions + 100 tasks across 10 real-world scenarios	Yes ¹⁹

¹ https://huggingface.co/datasets/takala/financial_phrasebank

² <http://people.eng.unimelb.edu.au/tbaldwin/resources/finance-sec/>

³ <https://sites.google.com/view/fiqa/home>

⁴ <https://github.com/yumoxu/stocknet-dataset>

⁵ <https://github.com/wuhuizhe/CHRRN>

⁶ <https://www.kaggle.com/datasets/daittan/gold-commodity-news-and-dimensions>

⁷ <https://github.com/czyssrs/ConvFinQA>

⁸ <https://github.com/SALT-NLP/FLANG>

⁹ <https://github.com/The-FinAI/PIXIU>

¹⁰ <https://github.com/SUFE-AIFLM-Lab/FinEval>

¹¹ <https://github.com/TongjiFinLab/CFGPT/tree/main/benchmark>

¹² <https://github.com/Duxiaoman-DI/XuanYuan>

¹³ <https://github.com/AlphaFin-proj/AlphaFin>

¹⁴ <https://github.com/YY0649/ICE-PIXIU>

¹⁵ <https://cfinbench.github.io/>

¹⁶ <https://github.com/snoels/fingeit>

¹⁷ <https://github.com/aliyun/cflue>

¹⁸ <https://github.com/aliyun/qwen-dianjin>

¹⁹ <https://github.com/FLAME-ruc/FLAME>

search, we identify and summarize seven key models that exemplify the current landscape of FinTSFM research. As illustrated in Fig. 3, these models can be broadly categorized into two types: those trained from scratch solely on time-series data, and those adapted from pre-trained language foundation models using both time-series and related textual or structured inputs.

3.1.1. Naive FinTSFMs Trained from Scratch

These models are pre-trained from scratch on financial time-series data using Transformer-based architectures. A representative example is MarketGPT [76], which uses a decoder-only Transformer trained directly on NASDAQ order flow data. It is designed to act as an order-generation engine within discrete event simulations. Another milestone

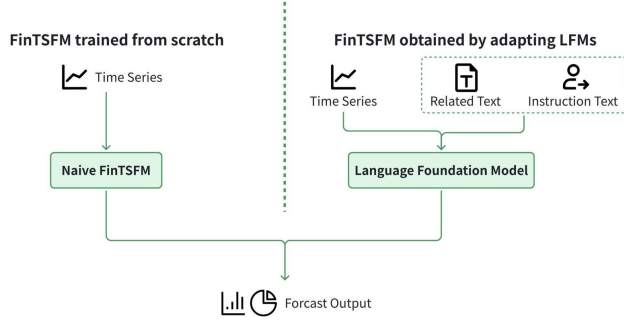


Figure 3: Two categories of FinTSFMs: models trained from scratch on time-series data (left) and models adapted from language foundation models using both time-series and related textual data (right).

in this category is TimesFM [77], a general-purpose time-series foundation model trained on real-world and synthetic time-series data in multiple domains including finance, transportation, weather, etc. Building on TimesFMs, several studies have adapted TimesFM for financial scenarios. For example, Fin-TimesFM [27] and FinDA-TimesFM [78] extend TimesFM via continual pre-training on stock, currency, and crypto time-series data to improve domain specificity. Other recent studies [79, 80] explore how timesFM can be adopted for value-at-risk and realized volatility forecasting. However, a limitation of naive FinTSFMs is that they typically handle only numerical time-series data, lacking multimodal capabilities (e.g., joint text-time reasoning), thus constraining their versatility in real-world financial scenarios.

3.1.2. FinTSFMs Adapted from Language Models

Large Language models (LLMs) are inherently designed for sequential data and have demonstrated strong cross-domain generalization due to their pretraining on large-scale web corpora. Their adaptation to time-series tasks is motivated by two key factors: (1) the efficacy of Transformer-based models in long-range sequence modeling [81, 82], and (2) their ability to integrate multimodal knowledge, including numerical reasoning [83, 84]. In this category, Time-LLM [85] introduces a reprogramming technique to convert time-series into natural language-like prompts, which are then fed into a frozen LLM using the Prompt-as-Prefix strategy for time-series reasoning. UniTime [86] extends this idea by integrating domain-specific instructions and multivariate time-series as input to a fine-tuned GPT-2 model, achieving competitive results. Furthermore, SocioDojo [87] offers a unique non-training approach, leveraging the agentic capabilities of GPT-3.5 and GPT-4. It combines structured knowledge bases, internet tools, and high-quality socioeconomic time-series to evaluate AI agents on a financial time-series value prediction task called “hyperportfolio”. Overall, FinTSFMs adapted from language models retain

the powerful language understanding capabilities of LLMs while expanding into the time-series domain, making them more suitable for broad and practical financial applications.

3.2. Training Method Analysis

We summarize the training strategies of representative FinTSFMs in Table 4. These models can be grouped into three categories based on their adaptation paradigm: time-series pre-training, training-based LLM adaptation, and non-training methods.

3.2.1. Time-series Pre-training

This strategy involves pretraining FinTSFMs directly on time-series data, typically using autoregressive objectives analogous to those used in language modeling. However, due to the continuous and multivariate nature of time-series inputs, these models often incorporate *patching* techniques [88] to divide input sequences into fixed-length segments. A representative example is TimesFM [77], along with its finance-specialized variants such as Fin-TimesFM [27] and FinDA-TimesFM [78], which employ patch-wise encoding to capture temporal dependencies across multiple variables. A particularly unique model in this category is MarketGPT [76], which tokenizes raw order-message data from electronic trading systems into discrete event sequences. This discrete simulation format enables autoregressive learning over market microstructure events, departing from typical numerical patching approaches and highlighting a distinctive design path.

3.2.2. Training-based LLM Adaptation

This approach adapts general-purpose language models to time-series tasks via additional training. A key distinction among existing methods lies in whether the LLM’s parameters are updated. For example, Time-LLM [85] preserves the frozen parameters of the LLM and introduces an adaptation layer that reprograms time-series segments into prompt-like inputs for reasoning. In contrast, UniTime [86] fine-tunes the LLM itself using multivariate time-series and domain-specific instructions, enabling it to directly model temporal dependencies. The contrast between these two approaches reflects the absence of a standardized adaptation pipeline for FinTSFMs—unlike FinLFMs, which increasingly follow a pretraining–SFT–alignment paradigm. This divergence suggests that training strategies for FinTSFMs remain exploratory and ripe for future formalization and innovation.

3.2.3. Non-training Methods

SocioDojo [87] exemplifies an alternative to training-intensive methods by leveraging large language models as reasoning agents without explicit time-series finetuning. Built upon GPT-3.5 or GPT-4, SocioDojo equips LLMs with access to external tools, including internet-based APIs, knowledge bases, and analytical modules. These agents interact with structured time-series inputs through multi-step planning and decision-making workflows, enabling tasks

Table 4
Training Methods of Representative FinTSFMs

Model	Backbone	Method	Training Data	Reference	Open Source
MarketGPT	Transformer	Time-series Pre-training	NASDAQ order flow data	[76]	Yes ¹
TimesFM	Transformer	Time-series Pre-training	Google Trends, Wiki stats, synthetic data	[77]	Yes ²
Fin-TimesFM	TimesFM	Time-series Pre-training	Stocks, indices, FX, crypto	[27]	Yes ³
FinDA-TimesFM	TimesFM	Time-series Pre-training	Stock price time-series	[78]	No
Time-LLM	GPT-2	Training-based Adaptation	Public time-series datasets	[85]	Yes ⁴
UniTime	GPT-2	Training-based Adaptation	Public time-series datasets	[86]	Yes ⁵
SocioDojo	GPT-3.5/4	Non-training	—	[87]	Yes ⁶

¹ <https://github.com/aaron-wheeler/MarketGPT>

² <https://huggingface.co/google/timesfm-1.0-200m>

³ https://github.com/pfnet-research/timesfm_fin

⁴ <https://github.com/KimMeen/Time-LLM>

⁵ <https://github.com/liuxu77/UniTime>

⁶ <https://github.com/chengjunyan1/SocioDojo>

such as hyperportfolio construction and dynamic forecasting. This approach underscores the potential of agentic LLM systems to reason over time-series data without extensive re-training, paving the way for lightweight and tool-augmented FinTSFMs.

3.3. Financial time-series-related Datasets

The commonly used datasets for financial time-series modeling is summarized in Table 5. These datasets vary in scale, format, and task coverage. For instance, the Google Stock Prices dataset [89] provides daily stock data from 2012 to 2017 and is widely used for basic market trend prediction. The S&P 500 dataset [90] offers a long historical span (1927–2020), supporting index-level forecasting and macroeconomic analysis. The Exchange Rate dataset [91], covering daily rates of eight currencies from 1990 to 2016, is frequently used to evaluate long-horizon forecasting models. The Bitcoin dataset [92] tracks cryptocurrency price fluctuations and serves as a benchmark for volatile financial assets. For multi-modal or multi-source financial applications, FNSPID [93] is a large-scale dataset integrating over 29.7M price records and 15.7M news headlines across more than 4,000 listed companies, enabling joint modeling of time-series, event impact, and sentiment effects. Most recently, FinTSB [94] has been introduced as a standardized benchmark for financial time series forecasting. It addresses key limitations of prior datasets—limited pattern diversity, inconsistent evaluation, and unrealistic trading assumptions. FinTSB consists of 20 datasets across four market regimes (uptrend, downtrend, volatility, black swan), each containing 300 stocks over 250 days. It offers unified metrics (ranking, portfolio, error), simulates real-world constraints (e.g., transaction fees), and provides detailed sequence characteristics (e.g., non-stationarity, forecastability).

Despite recent progress, the development of financial time-series datasets remains in its early stages. Most existing resources are constrained by limited scope or short temporal coverage, hindering comprehensive evaluation of long-context modeling, reasoning, and generalization. Continued efforts are needed to build larger, more diverse, and practically grounded benchmarks to support future advancements of FinTSFMs.

3.4. Summary

FinTSFMs represent a nascent yet rapidly evolving direction in financial AI research. While initial efforts have explored both from-scratch pretraining and adaptation from LLMs, a standardized modeling pipeline akin to FinLFMs has yet to emerge. Current methods span diverse training strategies and architectural choices, reflecting an exploratory landscape. Moreover, despite the availability of several high-quality datasets, benchmark development for long-context, multimodal, and reasoning-intensive financial forecasting remains insufficient. Continued advances in both model design and data ecosystem will be critical to unlocking the full potential of FinTSFMs in complex real-world financial applications.

4. Financial Visual-Language Foundation Models

This section focuses on Financial Visual-Language Foundation Models (FinVLFMs), which aim to jointly process financial visual information—such as line charts, candlestick diagrams, and scanned reports—and associated textual context to support tasks like visual question answering (VQA), document parsing, and multi-modal reasoning in financial domains. Similar to FinTSFMs, FinVLFMs are in early

Table 5
Common Financial Time-Series Datasets

Dataset	Scale	Task	Reference	Open Source
Google Stock Prices	Daily, 2012–2017	Stock price prediction	[89]	Yes ¹
S&P 500 Index	Daily, 1927–2020	Stock price prediction	[90]	Yes ²
Exchange Rate (8 Currencies)	Daily, 1990–2016	Exchange rate prediction	[91]	Yes ³
Bitcoin Prices	Daily, 2010–2020	Cryptocurrency trend forecasting	[92]	Yes ⁴
FNSPID (News + Prices)	1999–2023, 29.7M prices, 15.7M news	Integrated market analysis and price forecasting	[93]	Yes ⁵
FinTSB	Comprises 20 datasets, each with 300 stocks over 250 days, grouped into 4 distinct market patterns: uptrend, downtrend, volatility, and black swan.	Stock movement forecasting under realistic constraints	[94]	Yes ⁶

¹ <https://www.kaggle.com/datasets/vaibhavsn/google-stock-prices-training-and-test-data>

² <https://www.kaggle.com/datasets/henryhan117/sp-500-historical-data>

³ <https://github.com/laiguokun/multivariate-time-series-data>

⁴ <https://www.kaggle.com/datasets/soham1024/bitcoin-time-series-data-till-02082020>

⁵ <https://huggingface.co/datasets/Zihan1004/FNSPID>

⁶ <https://github.com/TongjiFinLab/FinTSBenchmark>

developmental stages, with only three representative works following relatively consistent design patterns. Our analysis introduces FinVLFMs through three lenses: (1) common architectural components, (2) training methodologies, and (3) specialized datasets that combine financial visuals (e.g., charts and report images) with textual content (e.g., news articles and regulatory filings).

4.1. Common FinVLFMs Architecture Components

As shown in Figure 4, most FinVLFMs follow a three-stage architecture: **Vision Encoder**: A pretrained image encoder (e.g., CLIP [95]) that transforms financial visuals into dense feature embeddings. **Vision Projector**: A lightweight adapter (e.g., Multi-Layer Perception, MLP) that aligns visual embeddings with the latent space of the LLM. **Base LLM**: A powerful language foundation model that integrates visual and textual inputs for downstream financial tasks.

In what follows, we detail each component individually and discuss how FinVLFMs may benefit from techniques developed in general-purpose VLMs.

4.1.1. Vision Encoder

The vision encoder converts diverse financial visual inputs—such as line charts, bar graphs, and tables—into embedding representations. Most existing FinVLFMs (e.g., FinLLaVA [19], FinTral [20]) employ CLIP or other general-purpose encoders. While convenient, this design overlooks the domain-specific visual patterns in finance. For example,

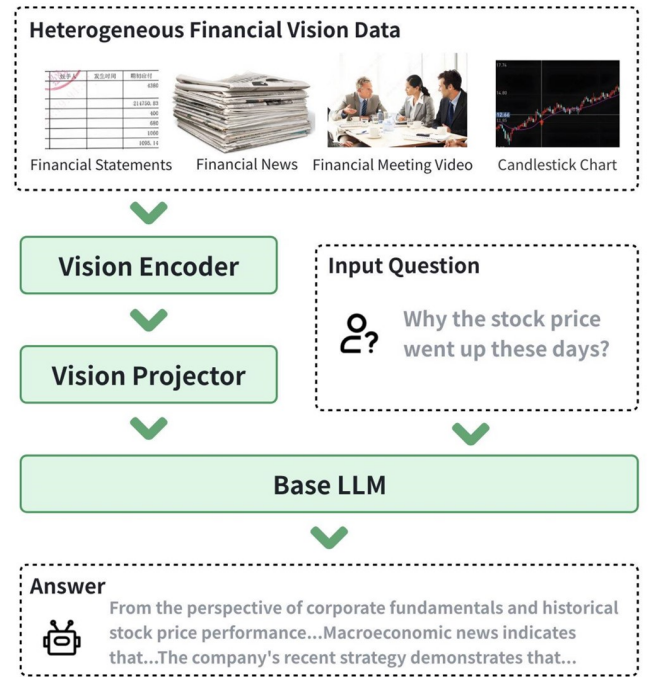


Figure 4: Current FinVLFMs share three common components: Vision Encoder, Vision Projector, and Base LLM.

line charts often require fine-grained resolution, and tabular data demands cell-level parsing—tasks not well-captured by generic visual encoders. Recent progress in general VLMs offers promising pathways. BLIP [96] improves image-text

Table 6
Training Details of FinVLFMs

Model	Backbone	Parameter	Training Method	Training Dataset Size	Reference	Open Source
FinVis-GPT	Vicuna	-	Modal Alignment PT, SFT	300K VQA pairs	[18]	Yes ¹
FinTral	Mistral	7B	Modal Alignment PT, SFT	1.86M VQA pairs	[20]	Yes ²
FinLLaVA	FinLLaMA	8B	Modal Alignment PT, SFT	1.43M VQA pairs	[19]	Yes ³

[PT]=Pre-training, [SFT]=Supervised Finetuning; [VQA]=Visual-Question-Answer.

¹ <https://github.com/wwwadax/FinVis-GPT>

² <https://github.com/UBC-NLP/fintral>

³ <https://www.thefin.ai/model/open-finllms>

alignment through bootstrapped contrastive learning, while LLaVA-UHD [97] enhances detail resolution using adaptive image patching. These methods can be directly adapted to improve the visual understanding capabilities of FinVLFMs.

4.1.2. Vision Projector

The vision projector aligns the visual embedding space with the LLM's token space, acting as the critical modality adapter. In current FinVLFMs, this layer is usually implemented using simple MLPs or linear projections (e.g., FinLLaVA [19] uses a 2-layer MLP). While sufficient for coarse-grained tasks, such basic aligners may struggle with complex visual semantics in financial reports. By contrast, general-domain VLMs increasingly employ cross-attention mechanisms [98, 99], gating layers, or dynamic adapters that better preserve inter-modal dependencies. Applying such techniques to FinVLFMs can enhance interpretability and context-aware reasoning, especially in tasks involving tabular-cell and chart-element mapping.

4.1.3. Base LLM

The base LLM decodes aligned multi-modal representations and performs financial reasoning. Early FinVis-GPT use Vicuna [100] as the backbone decoder. However, domain mismatch limits their financial fluency. To address this, FinLLaVA [19] introduced FinLLaMA, a LLaMA-based model further pretrained on financial corpora. Similarly, FinTral [20] leverages Mistral-7B for its superior numerical handling. These adaptations underscore the importance of integrating FinLFM in FinVLFM pipelines, particularly for tasks like earnings call summarization, portfolio optimization, or risk disclosure analysis.

4.2. Training Method Analysis

Table 6 summarizes representative FinVLFMs, including their model backbones, parameter scales, training methods, and dataset sizes. Despite architectural diversity, these models consistently follow a two-stage training paradigm: Modal Alignment Pre-training (PT) to align cross-modal

embeddings, and Supervised Finetuning (SFT) to enhance finance-specific reasoning.

4.2.1. Modal Alignment Pre-training

The first stage focuses on aligning the visual encoder and the language model within a shared semantic space. This is typically achieved by freezing the backbone components (i.e., the vision encoder and the LLM) and training the visual projection layers. All three FinVLFMs adopt this alignment-first approach, but differ in their alignment data construction strategies. FinVis-GPT [18] uses historical Chinese A-share market data to generate candlestick charts with technical indicators, supplemented by textual explanations generated by ChatGPT. FinTral [20] and FinLLaVA [19] integrates large-scale generic visual datasets with domain-specific financial data derived from stock price charts.

4.2.2. Supervised Finetuning

After modal alignment, the models undergo supervised finetuning to improve response quality and financial task alignment. Unlike the alignment stage, this phase often unfreezes both the LLM and the projection layer, enabling full parameter optimization. FinVLFMs differ in the granularity and content of their instruction datasets. Regardless of data source, the goal of this phase is to teach the model how to follow financial-specific prompts, interpret visual evidence, and generate coherent, grounded responses.

4.3. Financial Visual-Language-related Datasets

We summarize commonly used financial visual-language-related datasets in Table 7. These datasets cover a variety of modalities such as charts, tables, and documents. In the context of FinVLFMs, however, all non-text modalities—including tables—are typically rendered as images and processed through a unified visual encoder. This design simplifies the system pipeline and provides a consistent input format, which is beneficial for downstream financial applications.

As a result, most tasks across these datasets can be consistently categorized as Visual Question Answering (VQA),

regardless of the original modality. For example, datasets such as Australian and German Credit [101, 102] provide table-based VQA tasks targeting credit scoring scenarios. TAT-QA [103], FinQA [104], and ConvFinQA [105] focus on numerical VQA tasks grounded in financial earnings reports, often involving complex multi-step reasoning over visualized tabular data.

Chart-oriented datasets like ChartQA [106], FinVQA [20], and ChartBench [19] evaluate a model's ability to interpret line charts, bar plots, and other financial graphics for answering quantitative or descriptive questions. MME Finance [107] further expands this by integrating multiple financial visual formats—charts, tables, documents—into a unified evaluation suite for diverse VQA tasks.

In addition, MMMU Business [108], FCMR [109], and FAMMA [110] propose more complex, multi-modal or multi-hop financial VQA settings. These benchmarks introduce questions requiring reasoning over multiple types of inputs, integration of domain knowledge, and higher-level abstraction beyond single-modal analysis. Most recently, FinMME [111] collects over ten thousand of high-quality VQA with detailed annotation, aiming to establish a new benchmark for FinVLFMs.

Despite the diversity of content and design, most current datasets remain limited in scale, typically offering only hundreds to a few thousand VQA pairs. Thus, they are best suited for evaluating early-stage FinVLFMs within a multi-dimensional evaluation ecosystem and motivating the development of more robust, large-scale, and finance-specialized multimodal benchmarks. In particular, while these datasets are valuable for performance benchmarking, they are not sufficient to support the pretraining or large-scale instruction tuning of FinVLFMs. Future research will require the construction of significantly larger, more diverse, and task-rich financial visual-language datasets—encompassing real-world financial charts, tables, filings, and reports—to unlock the full potential of FinVLFMs.

4.4. Summary

FinVLFMs represent a promising yet nascent direction in financial AI, enabling multi-modal understanding across charts, documents, and textual narratives. Current models converge on a common three-component architecture and two-stage training paradigm, yet remain constrained by limited data scale, domain-specific visual encoding, and instruction diversity. To unlock the full potential of FinVLFMs in tasks such as financial auditing, investment reasoning, and automated reporting, future efforts must focus on scaling pretraining datasets, enhancing vision-language alignment, and integrating FinFLMs more deeply into visual pipelines.

5. FFM-based Financial Applications

In this section, we aim to review representative applications involving FFMs. Upon surveying the literature, we find that many existing studies still rely on general-purpose

foundation models (e.g., GPT-4, Gemini) to explore the feasibility and potential of FFMs in various financial tasks. However, several emerging works have begun to employ domain-specific FFMs (e.g., ICE-INTENT, TimesFM, CFGPT) that demonstrate clear advantages in bilingual understanding, compliance, and time-series prediction. Therefore, we select 11 representative application-oriented studies, encompassing both general-purpose and domain-specialized FFMs to illustrate the current landscape of FFM-based financial applications. We categorize these applications into four types: (1) **Financial Data Structuring**, (2) **Market Prediction**, (3) **Trading and Financial Decision**, and (4) **Multi-Agent Systems**. Table 8 summarizes representative studies within each category, where **domain-specific Financial Foundation Models are highlighted in bold**.

5.1. Financial Data Structuring

One of the foundational applications of FFMs lies in converting unstructured financial documents into structured representations, enabling downstream analysis and integration. In principle, all FFMs can be used as annotators or analyst for financial data. For example, Gu et al. [51] demonstrate that the domain-specialized model ICE-INTENT outperforms GPT-4 and other FinLFFMs in bilingual financial understanding. However, most early application-oriented studies still rely on general-purpose foundation models to initiate the structuring of financial data. For example, Aguda et al. [112] employ GPT-4 in a zero-shot setting to extract financial tuples (e.g., <organization, date>) for relation extraction. In the context of financial table parsing, researchers have explored the use of LLM to extract key items from corporate annual reports [113]. Together, these studies reflect a growing interest in leveraging both general-purpose and domain-specific FFMs to enable accurate and robust financial data structuring, laying the foundation for downstream analytical and predictive tasks.

5.2. Market Prediction

FFMs have also been increasingly applied to market forecasting tasks, such as predicting asset risk, market sentiment, and timing signals from financial text and time-series data. Recent work by Goel et al. [79] is among the first to apply a time-series foundation model (TimesFM) to Value-at-Risk (VaR) forecasting—estimating left-tail return quantiles for assets in the S&P 100 index. While such efforts highlight the emerging role of FinTSFMs, most current studies still rely on general-purpose LLMs to explore the feasibility of market prediction tasks in financial contexts. Fatouros et al. [114] develop MarketSenseAI, prompting GPT-4 with multi-source signals—including fundamentals, macroeconomic indicators, and news headlines—to generate investment rationales that outperform the S&P 100. Similarly, Chen et al. [115] propose a multimodal framework where GPT-4 evaluates investor forum comments to enhance sentiment-driven stock timing, improving profitability and win rates in backtests. Ko and Lee [116] investigate GPT-4's role in portfolio analysis, finding that it offers useful qualitative insights for stock selection and diversification when used alongside

Table 7
Common Financial Visual-Question-Answer (VQA) Datasets

Benchmark	Data Type	Task	Size	Reference	Open Source
Australian	table (credit records)	credit scoring	690	[101]	Yes ¹
German	table (credit records)	credit scoring	1000	[102]	Yes ²
TAT-QA	text, table	numerical VQA	19309	[103]	Yes ³
FinQA	text, table (earnings)	financial VQA	8281	[104]	Yes ⁴
ChartQA	chart	chart VQA	2500	[106]	Yes ⁵
ConvFinQA	text, table (earnings)	financial domain VQA	3892	[105]	Yes ⁶
MMMU Business	charts, diagrams, tables	business VQA	1428	[108]	Yes ⁷
FinVQA	chart (finance, stock market)	chart VQA	1025	[20]	Yes ⁸
ChartBench	chart	reasoning over charts	350	[19]	Yes ⁹
TableBench	table	reasoning over tables	450	[19]	Yes ¹⁰
MME Finance	charts, tables, docs	financial domain VQA	1171	[107]	Yes ¹¹
FCMR	text, table, image, chart	multi-hop VQA	2199	[109]	No
FAMMA	text, charts, tables	financial domain VQA	1758	[110]	Yes ¹²
FinMME	text, charts, tables	financial domain VQA	11099	[111]	Yes ¹³

¹ <https://archive.ics.uci.edu/dataset/143/statlog+australian+credit+approval>

² <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>

³ <https://nextplusplus.github.io/TAT-QA/>

⁴ <https://github.com/czyssrs/FinQA>

⁵ <https://github.com/vis-nlp/ChartQA>

⁶ <https://github.com/czyssrs/ConvFinQA>

⁷ <https://mmmu-benchmark.github.io/>

⁸ <https://github.com/UBC-NLP/fintral>

⁹ <https://chartbench.github.io/>

¹⁰ <https://tablebench.github.io/>

¹¹ <https://hithink-research.github.io/MME-Finance/>

¹² <https://famma-bench.github.io/famma/>

¹³ <https://huggingface.co/datasets/luojunyu/FinMME>

quantitative methods. These studies collectively show that while general-purpose LLMs dominate current explorations, financial foundation models—particularly FinTSFMs—are beginning to demonstrate unique advantages, paving the way for more specialized solutions in financial prediction.

5.3. Trading and Financial Decision

Beyond forecasting, FFM s are increasingly deployed in end-to-end financial decision-making pipelines, encompassing trading strategy development, investment advising, and risk control. For example, Li et al. [117] propose RA-CFGPT, a Chinese financial assistant built upon a fine-tuned FinLLM, a hybrid knowledge base under retrieval-augmented-generation framework, and compliance-aware checkers. The system demonstrates robust performance across tasks such as question answering, investment advising, and risk assessment. In other studies, general LLMs are explored for trading and financial decision. A representative work is FinMem, proposed by Yu et al. [118], which enhances LLM-based trading agents with a layered memory system and persona-driven architecture. By simulating short-, mid-, and long-term memory along with distinct investor styles (e.g., aggressive, conservative), FinMem achieves superior backtest performance compared to both deep reinforcement learning and vanilla LLM agents. Complementing this, Wang et al. [119] introduce LLMFactor, a sequential prompting framework that extracts interpretable alpha signals from financial news aligned with market

dynamics, improving both predictive performance and interpretability. Finally, Ko and Lee [116] demonstrate that GPT-4-guided portfolio construction can yield improved risk-adjusted returns, especially when combined with classical quantitative optimization. Collectively, these works illustrate the diverse ways in which FFM s can be adapted for high-stakes decision-making under uncertainty.

5.4. Multi-Agent Systems

Multi-agent financial simulations represent a promising frontier for FFM applications. Most current systems use general-purpose LLMs like GPT-3.5, GPT-4, or GPT-4o, but the underlying methods can naturally extend to FFM s. Xiao et al. [120] simulate realistic market interactions with agents assigned to trading-related roles (e.g., trader, risk officer), achieving coordinated decision-making through structured prompts. Lopez-Lira [121] simulates price formation and liquidity using GPT-4-driven market-making agents, offering new ways to test economic hypotheses. Similarly, StockAgent [122] emulates investor behavior in reaction to market news using multi-agent systems powered by GPT-3.5 and Gemini. While domain-specific FFM s are not yet widely adopted in this line of work, their incorporation could enhance realism and domain alignment in future agent-based simulations.

5.5. Summary

Current FFM-based financial applications predominantly leverage general-purpose models to explore feasibility across

Table 8

An Overview of FFM-Based Applications with Representative Works. Note that most studies adopt general-purpose foundation models for initial exploration; **domain-specific Financial Foundation Models are highlighted in bold**.

Task	Reference	Involved Models	Foundation	Description
Financial Data Structuring				
Bilingual Financial Text Understanding	[51]	ICE-INTENT, FinLLM, GPT-4	Disc-CFGPT,	ICE-INTENT supports 18 bilingual tasks and surpasses GPT-4 and other FinLLMs in bilingual financial understanding.
Financial Relation Extraction	[112]	GPT-4, PaLM 2, MPT		Zero-shot annotation of financial relations for efficient and reliable dataset creation.
Financial Table Parsing	[113]	GPT-4		Accurate extraction of values from annual report tables.
Market Prediction				
Value-at-Risk Forecasting	[79]	TimesFM		Evaluates TimesFM for left-tail Value-at-Risk prediction on S&P 100, showing superior performance over traditional models.
Stock Ranking from Multi-Source Signals	[114]	GPT-4		Outperforms S&P 100 with chain-of-thought investment signals.
Sentiment-Driven Trading Signal Enhancement	[115]	GPT-4		Improves timing by filtering investor sentiment.
Qualitative Portfolio Insight Generation	[116]	GPT-4		Assists stock selection and diversification with GPT-driven analysis.
Trading and Financial Decision				
Retrieval-Augmented-Generation-Based Financial Assistant System	[117]	CFGPT		Introduces a retrieval-augmented FinLLM with hybrid knowledge base, and compliance-aware checkers for tasks such as Question&Answering, investment advising, and risk control.
Trading with Memory and Personality	[118]	GPT-4		Introduces a layered memory architecture and persona-driven design to enhance trading decision-making.
Explainable Factor Extraction	[119]	GPT-4		Uses costumed prompting to extract interpretable market-moving factors from news.
Portfolio Diversification Support	[116]	GPT-4		Improves portfolio diversification and returns when combined with quant optimization.
Multi-Agent System				
Role-Specialized Trading Simulation	[120]	GPT-4o, OpenAI-o1		Multi-role AI agents collaborate to trade and manage risk.
Market Theory Testing via Agent Simulation	[121]	GPT-4		Multiple agents simulate realistic market to test financial theories.
Investor Behavior Emulation	[122]	GPT-3.5, Gemini		Multiple agents simulate different trading behaviors to real news and price data.

diverse tasks, including data structuring, prediction, trading, and simulation. However, emerging domain-specific FFMs (e.g., ICE-INTENT, CFGPT, DISC-FinLLM) are beginning to demonstrate advantages in financial understanding, compliance, and structured reasoning. These developments suggest a clear trajectory toward the integration of specialized FFMs, which are expected to improve accuracy, interpretability, and robustness in high-stakes financial environments.

6. Challenges and Opportunities

Despite the rapid progress in developing and applying FFMs, their widespread adoption still faces significant obstacles. These challenges arise not only from the unique characteristics of financial data and tasks, but also from the limitations of current algorithmic designs and computational infrastructure. Understanding and addressing these

bottlenecks is essential for advancing FFMs from proof-of-concept to practical deployment. To provide a structured perspective, we categorize the key challenges and corresponding research opportunities into three dimensions: **Data, Algorithm, and Computing Infrastructure**. Each dimension reflects a crucial aspect of the FFM development pipeline, from data acquisition to system-level deployment. In what follows, we highlight critical limitations, emerging solutions, and representative works that pave the way for future progress.

6.1. Data

6.1.1. Scarcity of Large-Scale Multimodal Financial Datasets

As discussed in Sections 2–4, current FFMs—especially FinVLLMs and FinTSFMs—are constrained by the lack of large-scale, high-quality multimodal datasets (e.g., chart-text or table-report pairs). Researchers have highlighted the

issue of scarce corpora and underdeveloped multimodal capabilities in FFM [19], limiting their real-world applicability. However, acquiring such data from real financial environments is costly and heavily regulated. An alternative strategy is *data synthesis* using advanced LLMs, which is emerging as a promising approach to construct financial benchmarks. For example, He et al. [123] generate synthetic chart reasoning datasets by programmatically producing visual plots and pairing them with high-quality QA items. The automatic generation of accurate, diverse financial data remains a key research direction.

6.1.2. Privacy and Data Confidentiality

Many valuable financial datasets—such as internal bank reports, client transactions, and proprietary trading strategies—are confidential and inaccessible to the public due to privacy regulations and competitive concerns. This severely limits data sharing and aggregation for FFM training. A promising solution lies in *federated learning*, which allows multiple institutions (e.g., banks, hedge funds) to collaboratively train models without sharing raw data. For general-domain LLMs, several works [124, 125] have already explored federated training paradigms that preserve data privacy. These approaches offer valuable insights and technical foundations for privacy-preserving FFM development.

6.2. Algorithm

6.2.1. Hallucination and Factual Inconsistency

Foundation models are prone to hallucination—generating incorrect or fabricated information. In financial contexts, such errors can have severe consequences, particularly when models produce false statements regarding earnings, asset risks, or regulatory events. Even state-of-the-art systems like GPT-4 have been shown to generate unverified or inconsistent financial claims [111]. Addressing this issue requires integrating FFMs with structured knowledge sources. For instance, Jiang et al. [126] propose a multimodal Retrieval-Augmented Generation (RAG) system that combines a vector database and a financial knowledge graph. By converting visual data into text and retrieving structured facts across modalities, the system supports LLM reasoning grounded in reliable information. Such integration between FFMs and knowledge engineering offers a promising path toward trustworthy financial applications.

6.2.2. Lookahead Bias in Financial Backtesting

FFMs are typically trained on large corpora that may contain future information relative to backtest periods, introducing *lookahead bias*. This leads to inflated performance during model evaluation because the model has inadvertently learned about future events. For example, a foundation model evaluated on data from 2020 may have been trained on news published after 2020, violating the temporal integrity of the backtest [127]. To mitigate this, careful curation of temporally consistent datasets is essential. One example is TimeMachineGPT [128], where models are trained exclusively on data constrained to specific time horizons to prevent future leakage. Developing temporally aware training

and evaluation protocols remains a critical direction for FFMs.

6.3. Computing Infrastructure

6.3.1. High Training and Deployment Costs

Training or fine-tuning FFMs with tens of billions of parameters is extremely resource-intensive, often requiring thousands of GPUs, extensive memory, and weeks of compute time. This presents a significant financial and infrastructural barrier, especially for academic institutions and small enterprises. For example, BloombergGPT [15] was trained using approximately 1.3 million GPU hours on NVIDIA A100s—an effort estimated to cost \$1–2 million. In addition to training costs, deploying large-scale FFMs also poses substantial challenges. Inference with large models typically incurs high latency, substantial energy consumption, and complex infrastructure requirements, particularly in real-time or low-latency financial applications such as trading, risk monitoring, or customer interaction systems. In response, research has begun to explore lightweight alternatives such as distilled, small yet powerful FFMs derived from general-purpose LLMs [9], which aim to retain financial domain capabilities while significantly reducing computational demands. Beyond simple model compression, an emerging paradigm advocates for a collaborative system that integrates large and small models. Researchers [129, 130] discuss hybrid systems where large LLMs are used to provide domain knowledge, supervision, or feature augmentation, while lightweight models handle privacy-sensitive or latency-critical tasks on the edge. This collaboration balances performance, privacy, and efficiency—making it particularly attractive for financial scenarios with strict regulatory or cost constraints. Therefore, future FFMs may no longer be monolithic; instead, they are likely to evolve into modular systems composed of both large and small models working in synergy.

7. Conclusion

FFMs are reshaping the landscape of financial engineering by enabling scalable, adaptable, and multi-modal intelligence across tasks such as analysis, forecasting, and decision-making. In this survey, we systematically reviewed the architectural designs, training methodologies, benchmark datasets, and practical applications of FFMs, categorizing them into three main types: FinLFMs, FinTSFMs, and FinVLFMs. Despite notable progress, FFMs still face a range of open challenges, spanning from data availability and temporal consistency to model trustworthiness, efficiency, and domain adaptation. Tackling these issues will require advances across multiple fronts, including high-quality dataset construction, multi-modal integration, efficient training and deployment strategies, and deeper alignment with financial knowledge and regulatory requirements. We hope this work provides a useful foundation for further research and encourages the responsible development of FFMs in real-world financial systems.

CRedit authorship contribution statement

Liyuan Chen: Conceptualization of this study, Methodology, Writing - Original draft preparation, Project administration. **Shuoling Liu:** Conceptualization of this study, Methodology, Writing - Original draft preparation. **Jiangpeng Yan:** Conceptualization of this study, Software, Methodology, Writing - Original draft preparation. **Xiaoyu Wang:** Data curation, Writing - Original draft preparation, Visualization. **Chuang Li:** Data curation, Writing - Original draft preparation. **Henglin Liu:** Data curation, Writing - Original draft preparation. **Kecheng Jiao:** Data curation, Writing - Original draft preparation. **Jixuan Ying:** Data curation, Writing - Original draft preparation. **Yang Veronica Liu:** Writing - Review & Editing, Supervision. **Qiang Yang:** Writing - Review & Editing, Supervision. **Xiu Li:** Writing - Review & Editing, Supervision.

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